AN APPROACH TO FEATURE EXTRACTION OF HUMAN MOVEMENT QUALITIES AND ITS APPLICATION TO SOUND SYNTHESIS

Mark Pedersen

School of Performing Arts
Victorian College of the Arts and Music
University of Melbourne

Roger Alsop

School of Performing Arts
Victorian College of the Arts and Music
University of Melbourne

ABSTRACT

This study investigates qualities of human movement that relate to the artistic organisation of sound. Identification of human movement qualities can assist in making expressive computer music interfaces and sound art installations which respond appropriately to expressive human movement.

In this paper, we present an empirical approach to exploring the relationship between sound and movement, in which participants are asked to move in response to sounds selected randomly from a palette of samples based on a mapping between Laban movement qualities and spectromorphological properties. Participants' movement is recorded using a depth-sensing computer vision system and later analysed with respect to the movement qualities exhibited. Participants are also asked to verbalise their thoughts and responses in a follow up interview.

The collected data, will be used in the design of an interactive installation based on the principles of the relationships between sound, movement and cognition established through our investigation.

1. INTRODUCTION

The aim of the research is to establish an understanding of common associations between organised sounds and movement qualities in order to facilitate the creation of expressive computer music interfaces, responsive art installations, and soundscapes for contemporary dance theatre.

The significance of this research is evidenced by the growing interest in computer music interfaces, seen in the New Interfaces for Musical Expression conference and in the burgeoning commercial market for computer music instruments and broader movement-focused computer game interfaces such as the Nintendo Wii and the Microsoft Kinect.

Our hypothesis is that, within a given local culture, a common set of associations between sound qualities and movement qualities may exist and such an association is bi-directional and symmetric (i.e. that specific sound qualities will evoke specific movement qualities in response, and vice versa). There is an underlying assumption that specific movement qualities are expressive of a person's expressive intent.

We aim to benefit the creative community by informing the design of both artistic works and commercial products which depend upon the automatic interpretation of user intent through physical movement, particularly where that interpretation results in the production of sound or music, or where sound or music is used to shape human responses to an environment. As an example, Maes et. al [1] have recently published work on a gestural control interface for vocalists.

2. RELATED WORK

This work builds upon a strong body of literature, well summarised in Godøy and Leman [2]. For example, Hevner's pioneering work in establishing empirical connections between emotion and music [3] [4] [5] helped to establish an experimental framework for investigating the psychological basis for human responses to music, and this approach has underpinned the majority of subsequent studies. In collecting a statistically significant set of response data Schubert et al. [6] shows that there are common emotional responses to musical features, and that the subjects' felt response is more significant than the emotion that subjects perceived was intended to be expressed by the music [7] [8].

For this reason, capturing subjective responses to sound and movement form an important part of this research. Inclusion of these subjective responses will also form an important part of the final installation as a bridge between observable stimuli and internal response, and to allow participants in the final artwork to gain insight into the experience of others at a cognitive and linguistic level.

Like Maes [1], we approach this work on the basis of the Embodied Music Cognition paradigm [9]. This perspective provides both empirical evidence and an explanatory theory which allows for a basic level of "objective" meaning to inhere in sound due to humanity's common cognitive strategies, while still allowing for individual and cultural differences. On this basis, we consider an investigation into associations between movement qualities and sound qualities will make a useful contribution to the overall understanding of embodied cognition of sound, as well as having practical application to the design of expressive music performance interfaces.

3. APPROACH

In examining the relationship between movement and sound, we have taken an empirical approach similar to that used by Maes [1], in which we observe spontaneous movement in response to sound stimuli. The following sections describe the design and selection of the sound stimuli, the movement capture system, and the analysis and results.

3.1. Movement Qualities and Sound Design

Central to our hypothesis is the idea of a bidirectional relationship between sound and movement qualities, arising from the premise that, as embodied beings, our sound processing capacity is related to the way we generate and perceive movement at the neurological level.

Drawing upon Laban Movement Analysis (LMA) [10] we sought to identify correlations between Laban Effort Shapes, as a canonical set of movement qualities, and the spectromorphological attributes of sound developed by Smalley[11]. Marshall[12] takes a similar approach in the application of Smalley's spectromorphology to the design of an interactive sensor-based sonification system. Smalley's theory of source-bonding is applied to the design of sounds to exploit participant's natural tendency to associate sounds with a (potential) source. Principles of Auditory Scene Analysis are also applied to ensure that individual auditory streams remain distinct and recognisable.

In our approach, the concept of source bonding is explored through empirical observation of the participants' spontaneous movement in response to a variety of sound stimuli drawn from Smalley's spectromorphological typology. The association between physical gesture and spectromorphological type are inherent in that Smalley's typology and LMA effort shapes share several identical or highly similar categories, such as floating, pushing/pressing, flinging/flicking. Leman's work on embodied musical cognition suggest that these associations are grounded in the very cognitive processes used for the perception of sound. Godøy[17] further proposes an explicit linkage between Schaeffer/Smalley's typology/morphology and sound producing/accompanying gesture.

Other associations between Smalley's typology and LMA are not so obvious at the category level, and we need to examine common combinations of qualities in order to identify a mapping.

First, we consider the components of effort in LMA, which are Space, Weight, Time and Flow. Each of these forms an axis in a four dimensional model of movement effort, as shown in Figure 1.

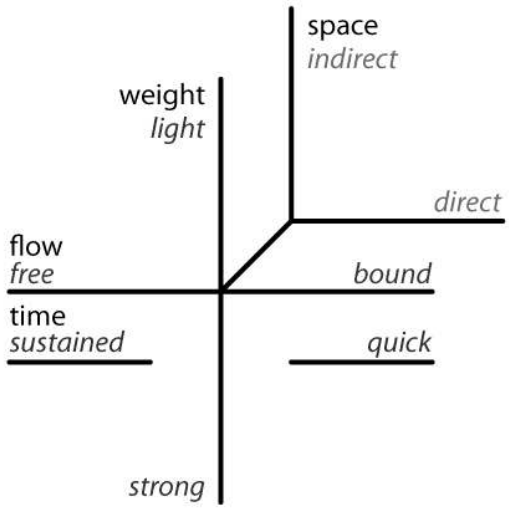


Figure 1. Laban Effort Graph [16]

The dimensions of the Laban Effort Graph are well summarized by Zhao[13], as shown in Table 1. The Space, Time and Weight factors of the effort model produce eight basic effort types, summarised in Table 2. Flow is considered separately, as explained by Laban[14]: "Movements performed with a high degree of *bound flow* reveal the readiness of the moving person to stop at any moment in order to readjust the effort if it proves to be wrong, or endangers success. In movements done with *fluent flow*, a total lack of control or abandon becomes visible, in which the ability to stop is considered inessential.

Motion factor	**Dimensions**	**Examples**
Space attention to the surroundings	**Indirect (flexible)**: spiraling, deviating, flexible, wandering, multiple focus **Direct**: straight, undeviating, channeled, single focus	Waving away bugs, surveying a crowd of people, scanning a room for misplaced keys Threading a needle, pointing to a particular spot, describing the exact outline of an object

Weight attitude to the movement impact	**Light**: buoyant, weightless, easily overcoming gravity, marked by decreasing pressure **Strong**: powerful, forceful, vigorous, having an impact, increasing pressure into the movement	Dabbing paint on a canvas, pulling out a splinter, describing the movement of a feather Punching, pushing a heavy object, wringing a towel, expressing a firmly held opinion
Time lack or sense of urgency	**Sustained**: leisurely, lingering, indulging in time **Sudden (quick)**: hurried, urgent, quick, fleeting	Stretching to yawn, striking a pet Swatting a fly, lunging to catch a ball, grabbing a child form the path of danger, making a snap move
Flow amount of control and bodily tension	**Free (fluent)**: uncontrolled, abandoned, unable to stop in the course of the movement **Bound**:	Waving wildly, shaking off water, flinging a rock into a pond Moving in slow motion, tai chi, fighting back tears, carrying a cup of hot tea

Table 1. Motion Factors and Effort Elements (from Zhao 2001, p. 41)

In terms of spectromorphology, Smalley[11] describes seven characteristic motions: fly, float, drift, throw/fling, rise, flow, push/drag, all of relate to dimensions of *motion rootedness*, *motion launching (attack), contour energy and inflection, and internal texturing.* Note that these are motions in terms of the spectra of sound, rather than motions in spatialisation of sound. The relationship between these dimensions and the characteristic motions is not fully explained by Smalley, however indications are given in the text and summarized in Table 3.

Effort Type	**Space**	**Weight**	**Time**
Slashing	Flexible	Strong	Quick
Gliding	Direct	Light	Sustained
Pressing	Direct	Strong	Sustained
Flicking	Flexible	Light	Quick
Wringing	Flexible	Strong	Sustained
Dabbing	Direct	Light	Quick
Punching	Direct	Strong	Quick
Floating	Flexible	Light	Sustained

Table 2. Basic Laban Effort Types

Motion type	**Rooted - ness**	**Attack**	**Contour**	**Texture**
Fly	rootless	emerging	slow / evolving	sustained
Float	rootless	emerging	slow / evolving	sustained
Drift	rootless	emerging	slow / evolving	sustained
Throw/ Fling	partial	pressured	rapid	granular
Rise	partial	emerging / pressured	rapid	streaming / turbulent
Flow	rooted	emerging / pressured	rapid	sustained / granular
Push/ Drag	rooted	pressured	slow	turbulent

Table 3. Spectromorphological motion qualities

In comparing Tables 2 and 3, it is possible to derive some correlations between the two models: *time* in LMA relates to both the *motion launching (*or *attack)* and the *texture* quality of spectromorphology; *weight* relates to the *rootedness* quality; and *space* relates to *contour*. This mapping is imperfect, but is sufficient for us to design the sounds which form the basis of our empirical approach to examining the relationship between movement and sound. Expressing Smalley's spectromorphological movement qualities in terms of more common synthesis parameters, we have taken the following approach to designing/selecting sounds which we expect to correspond to various LMA movement qualities. These are summarized in Table 4.

Motion Factor	**Movement Quality**	**Synthesis properties**
Time	Quick	short percussive sounds (short onset, short continuation)
	Sustained	Sustained sounds (longer onset, long continuation, long decay)
Weight	Light	low density sounds with harmonic content and greater reverb, higher pitch
	Strong	high density sounds with greater noise, less reverb, lower pitch
Space	Direct	steady pitch and spectra, less reverb
	Flexible	varying pitch and spectra, more reverb

Table 4. LMA Motion Factors and sound synthesis properties

4. DATA COLLECTION METHOD

At the time of writing, data collection and analysis is on-going, with 5 participants so far. Participants we recruited by referral and were screened for basic aptitude with respect to the tasks of the study. Participants come from a mixture of backgrounds, with 60% having greater than 5 years music performance training and experience. All participants reported that they listened to music for recreation and to alter their mood.

Sounds were designed or selected from existing corpora according to the qualities described above and assembled into a playback system that allowed for random selection and playback, one sample at a time. Participants were asked to respond to the sounds they heard by moving in whatever way they felt appropriate. The nature of the study does not lend itself to the use of a control group, however participants were given a trial task to confirm their understanding of the task and resolve any questions or concerns.

The order of the triggered sounds was recorded in a sequencer for later alignment with movement data. Movement data was collected using a Microsoft Kinect sensor and software from OpenNI [15] which allows RGB and depth-sensing camera data to be recorded. Participants were then interviewed and audio from the interview session was recorded[1].

5. ANALYSIS

5.1. Movement Analysis

Initial investigations into the data was conducted by manual tagging of the audio/video stream, labeling sections of video with instances of the recognition methods shown in Table 5.

Labelled audio/video data was then segmented according to the sound stimulus, and aligned across all subjects, allowing us to identify common movement qualities that emerged for any given stimulus (e.g. sustained movement for sustained sound, e.g. floating).

Basic Effort Type	Recognition Method
Gliding (sustained, direct, light movement)	long duration, low velocity movement without changes in direction, maintaining distance between hands and torso.
Slashing (quick, flexible, strong)	long duration, high velocity movement with changes in direction
Punching (quick, direct, strong)	short duration, high velocity movement with suddent acceleration and deceleration involving whole arm (elbow moves from torso)
Floating (sustained, flexible, light)	long duration, low velocity movement with changes in direction
Dabbing (quick, direct, light)	short duration, low velocity move-ment
Pressing (sustained, direct, strong)	long duration, low velocity movement without changes in direction, hands moving away from the body
Flicking (quick, flexible, light)	short duration, high velocity movement localised to the hands (elbow doesn't move far from torso)
Wringing (sustained, flexible, strong)	long duration, low veloc- ity movement with hands constrained body space (hands and arms close to torso)

Table 5. Effort type recognition methods

Table 6 shows the results for data collected to date. Shaded cells indicate the intended quality criteria for a given effort type, while number values give the normalised number of actual observations of that quality criteria for the given sound stimuli category.

Some sound stimuli were consistently associated with the expected movement criteria. Sounds in the floating and wringing categories produced expected behaviour; gliding also produced fairly consistent results, with some variation velocity and constancy of hand-torso position. Dabbing and flicking responses were crossed over in their observed velocity, but tended in the correct directions (lower for dabbing, higher for flicking): quantitative analysis of the motion capture data will be used to more closely investigate these responses to determine whether a clear boundary can be observed in terms of velocity. Some stimuli, such as wringing, tend to be variable in the movement responses they generate. Others, such as slashing and punching, generate visibly different movements, which are not strongly distinguished by the recognition criteria, and do not align with the expected results.

5.2. Analysis of interviews

Post-activity interviews with subjects revealed a number of interesting threads. A common theme was that subjects moved in response to visual imagery, i.e. their imagination of how the sound looked, rather than how they felt in response to the sound. Subjects reported that they had little emotional response to the sounds, and moved in a way that mimic'd the sound, rather than expressing personal intent.
Some subjects reported that complex or ambiguous sounds offered more room for interpretation and

[1] This research is being conducted under ethics approval for Minimal Risk Application 1136360.1, University of Melbourne.

personal expression, compared to sounds which were coherently of one type (e.g. percussive sounds or floating sounds).

Other common observations were the relevance of certain sounds to parts of the body. Subjects commonly related higher pitched or short duration sounds to the extremities, while lower pitched sounds with rich harmonic content were related to the whole body.

6. CONCLUSIONS AND FUTURE WORK

Our objective in examining the sound-movement mapping is to ultimately reverse mapping and use movement quality recognition to control various sound synthesis parameters, and in this way, build a gestural control system which is expressive of the users intent.

The stimulus-response data collected and the subject observations collected so far are encouraging in terms of supporting the underlying embodied music cognition approach that we have taken. The specific movement quality attributes selected are generally robust in terms of categorising the observed overall movement quality, and support the idea that there is an association between sound qualities and movement qualities that manifest in observably common ways. Refinement of the selected quality criteria for certain sound stimuli is needed to capture the wider variety of movement observed.

The most unexpected result observed so far is that our assumption that this association is bi-directional and symmetric appears to be false. Subjects report that, while their movements related to shape of the sounds, they were, in general, not expressing any intent. Even though subjects were clearly in control of their own bodies, their comments indicate a general sense of lacking agency.

This was particularly evident in one case, where in response to sounds designed to relate to the 'punching' effort type, the subject moved *as if being punched*. If the subject were to be moving with the intention of generating a punching sound, it would be more natural to be moving as if punching (in line with the general embodied music cognition theory that sound is often processed in terms of what motion would produce that sound).

In retrospect, the asymmetry of the agent-patient relationship is obvious, but it does raise interesting questions about what issues will arise when attempting reverse the sound-motion mapping developed during this phase of our research and apply it to synthesis control. Even more interesting, however, is the subjects' reported sense of lack of intent in responding to sound stimuli. This raises many questions in terms of the way in which interactive synthesis systems may be used to mediate performer-audience, performer-instrument, and performer-performer relationships, and is an area which we intend to explore further.

In terms of practical future work, additional data collection and analysis is required to gain greater confidence in the results we have presented here, as well as gain greater insight into the most effective movement quality criteria for effort type recognition.

The manual motion quality analysis conducted in this study will be used to build an automated motion quality recognition system utilising the OpenNI skeleton tracking system to extract 3D coordinates of key joints (hands, elbows, head, torso, hips, knees, ankles) and analyse the data according to the motion quality recognition methods presented here.

In addition, a supporting study into classification of the designed sounds is also planned. Subjects will be requested to apply their own freely associated labels to sounds they have grouped together, as well as applying the fixed set of effort type labels in order to validate the sound-movement association at a cognitive labelling level.

The understanding gained through completing these studies will be used to drive the development of a gestural control system for synthesis, to be used in an interactive audio-visual installation. We also expect to be able to apply this deeper understanding of sound-movement

Movement qualities	Pressing	Wringing	Dabbing	Punching	Floating	Gliding	Slashing	Flicking
Long duration	1.00	1.00	0.00	0.50	1.00	1.00	0.38	0.00
Short duration	0.00	0.00	1.00	0.50	0.00	0.00	0.63	1.00
Low velocity	1.00	0.80	0.25	0.75	1.00	0.67	0.13	0.00
Med velocity	0.00	0.20	0.75	0.25	0.00	0.17	0.38	0.67
High velocity	0.00	0.00	0.00	0.00	0.00	0.17	0.50	0.33
Steady direction	0.00	0.00	0.75	0.25	0.00	0.33	0.75	1.00
Changing direction	1.00	1.00	0.25	0.75	1.00	0.67	0.25	0.00
Sudden accel/decel	1.00	0.40	0.75	1.00	0.00	0.17	1.00	1.00
Gradual accel/decel	0.00	0.60	0.25	0.00	1.00	0.83	0.00	0.00
Hand-torso distance constant	0.00	0.00	0.00	0.25	0.25	0.50	0.00	0.00
Hands-torso distance increasing	0.00	0.00	0.50	0.25	0.63	0.50	0.63	1.00
Hand-torso distance variable	1.00	1.00	0.50	0.50	0.13	0.00	0.13	0.00
Elbow-torso distance constant	0.00	0.20	0.50	0.25	0.25	0.50	0.00	0.00
Elbow-torso distance increasing	0.00	0.00	0.25	0.25	0.63	0.33	0.75	1.00
Elbow-torso distance variable	1.00	0.60	0.25	0.50	0.13	0.17	0.25	0.00

Table 6. Observed movement quality criteria for various sound stimuli (normalized average number of instances).

association to sound design for dance performance, and computer game audio in way that delivers inherently meaningful audio responses to performer/user gesture while going beyond simple mappings which lack nuance and lasting appeal.

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